IEG Wafer Fab SOP Trainer — Instructor Script Guide (Front-End and Back-End)

IEG Co., Ltd. · Global Business Division · Digital Twin Suite resource 10 · Standard reference data edition (v0.2)

Revision v0.2: the metallisation lesson now separates the wafer pre-clean from the target pre-sputter and adds a contact-resistance measurement. Simulator references use the STEP and MODULE numbers of the 6D simulators, and the failure shapes follow the 6D instructor guides.

1. How to Use the Trainer

The IEG Wafer Fab SOP Trainer lets a student perform one fabrication process step by step, in the correct order, with the correct settings and safety checks. This guide gives a lesson script for each of ten processes: five front-end (lithography, etching, ion implantation, CVD, metallization) and five back-end (die bonding, wire bonding, molding, saw/dicing, package testing).

Data basis. All recipes use standard reference values typical of textbooks and teaching labs, not the settings of any specific lab. Process results come from a simplified training model that shows the right direction and size of effects, not a calibrated simulation. Equipment-specific recipes can be loaded later (Appendix).

Training modes

Mode	What the student sees	Use it for
Guided	Next action highlighted, target values shown and prefilled, a short explanation for each step	First exposure; teaching the sequence
Practice	No prefilled values; out-of-sequence actions are blocked with a message; each setting is graded after it is applied	Rehearsal and what-if experiments
Assessment	No hints and no feedback; errors are recorded silently; a report appears at the end	Marked evaluation

Scoring (100 points)

Component	Weight	How it is scored
Sequence	40	Each out-of-sequence action costs 15 % of this component
Settings and decision	40	Each setting: full credit within tolerance, half within twice the tolerance, none beyond; the pass/rework decision counts as one more item
Safety	20	Each safety step skipped or done too late costs 34 % of this component

Bands: 85 and above is Competent; 70–84 needs coaching; below 70 repeat the procedure. A wafer that misses its specification is capped at 84, so a student cannot be rated Competent after producing a failed wafer.

Work orders (randomised targets)

Every run starts with a work order that changes the target values, so students cannot memorise one set of numbers or pass answers to each other. The work order appears in a card at the top of the screen. The student reads the target and the reference data, then calculates or looks up the settings.

- **Work order menu: Random** *each run* (default; never repeats the previous order), *Standard* (the original values), or a specific order. Use a specific order when you want the whole class on the same case, or to give each student a different named order in an assessment.
- **Modes:** Guided shows target values and prefills them, so use Practice or Assessment when students must calculate.
- **Reports:** the work order ID is printed on the report and in the result file, so you can check each student's case.

Process	What the work order varies	What students calculate or read	Example
Lithography	Target resist thickness 0.8–1.5 µm	Spin speed = 4,000 rpm ÷ thickness^2	0.8 µm needs 6,250 rpm
Etching	Oxide thickness 400–800 nm	Etch time = thickness × 1.1 ÷ 100 nm/min	800 nm needs 528 s
Ion implantation	Dose 0.5–2.0×10^{15}, beam 1–2 mA, energy 40–60 keV	Time = Q·q·A ÷ I	2.0×10^{15} at 1.0 mA needs 57 s
CVD	Oxide thickness 300–1,000 nm	Time = thickness ÷ 200 nm/min	1,000 nm needs 300 s
Metallization	Aluminium thickness 600–1,200 nm	Time = thickness ÷ 600 nm/min	600 nm needs 60 s
Die bonding	Die size 1.5–3.0 mm and epoxy cure schedule	Dispense time and force scale with die area; cure from the datasheet	3.0 mm die: 270 ms, 225 gf
Wire bonding	Wire diameter 18–30 µm	Force and power scale with diameter squared; loop is 6 × diameter	18 µm wire: 26 gf, 41 mW, 108 µm
Molding	EMC datasheet A–D	Read mold temperature, pressure, transfer and cure time	EMC-B: 170 °C, 6 MPa, 15 s, 100 s
Saw / dicing	Blade and package datasheet A–D	Read spindle, feed, coolant and cut depth	Thick package: 35 krpm, 40 mm/s
Package testing	Test spec A–D	Read voltage, temperature, test time and contact force	Hot test: 1.8 V, 85 °C

Note on the lessons. The what-if numbers in Parts 2–7 are for the Standard work order. With any other order, have students predict relative changes instead (for example, "20 % more dose"), and check the result on screen.

Standard 120-minute lesson model

Minutes	Block	What happens
0–10	Safety hook	Instructor tells a short real-world safety story for the process
10–30	Concept script	Instructor explains the physics and the SOP logic using the script in Parts 2–7
30–60	Guided run	Students perform the full SOP in Guided mode
60–85	What-if exercises	In Practice mode, students change one setting and predict the result first
85–110	Assessment run	Each student performs the SOP in Assessment mode
110–120	Debrief	Review reports; connecting findings to the 6D simulator Fault Cases

Script conventions

- **SAY:** words to speak, close to verbatim.

- **DO:** an instructor action.
- **ASK:** a question for the class, with the expected answer.
- **WATCH:** a mistake or misconception to listen for.

Classroom setup

☐ One desktop or laptop per student or pair; a current browser; no internet needed once the file is open.

☐ The file `IEG_Wafer_Fab_SOP_Trainer.html` on every device, or hosted with the Digital Twin Suite hub.

☐ Projector for the instructor demonstration.

☐ Choose the process in the **Process** menu and the case in the **Work order** menu at the top; **Restart** resets the session; **Recipe data** loads or exports recipe files.

☐ Students keep a paper or digital logbook: setting changed, prediction, result, lesson.

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2. Lithography SOP Lesson

Students coat, bake, expose and develop one wafer in nine steps and are scored on sequence, settings and safety. Pair this lesson with Front-End STEP 03 Photolithography in the 6D simulator curriculum.

Objectives. Students can (1) perform the lithography SOP in the correct order, (2) explain why each bake, dose and time matters, (3) predict how a wrong setting changes thickness or linewidth.

Standard recipe and specification

Step	Setting	Standard value	Tolerance
Prime	Dehydration and prime temperature	120 °C	± 10
Prime	Prime time	60 s	± 15
Coat	Dispense volume	3 mL	± 1
Coat	Spread speed	500 rpm	± 150
Coat	Spin speed	4,000 rpm	± 300
Coat	Spin time	30 s	± 5
Soft bake	Temperature and time	100 °C, 60 s	± 5 °C, ± 10 s
Expose	Dose	150 mJ/cm²	± 15
Expose	Focus offset	0 µm	± 0.3
Post-exposure bake	Temperature and time	110 °C, 60 s	± 5 °C, ± 10 s
Develop	Develop time and rinse time	60 s and 30 s	± 10 s each
Hard bake	Temperature and time	120 °C, 120 s	± 10 °C, ± 30 s

Specification: resist thickness 1.00 ± 0.05 µm; space width (critical dimension) 1.00 ± 0.10 µm. Standard settings give 1.00 µm and 1.00 µm.

Script

[0–10] Safety hook.

- **ASK:** "This bay has a hot surface, a corrosive liquid, flammable vapour and UV light. Which is the most likely to hurt someone who is in a hurry?" **Expected:** accept any answer with a reason; steer to hot surfaces (hotplate) and the corrosive developer as the everyday risks.
- **SAY:** "Every safety step in this trainer is worth marks, and doing it too late counts as skipping it."

[10–30] Concept script.

- **SAY:** "Lithography is photography for circuits. The resist is the film, the mask is the negative, and the UV light is the exposure. Everything after this station copies what happens here."
- **SAY:** "Spin speed sets thickness: faster spin, thinner film, roughly with the inverse square root of the speed. Dose sets how much resist changes. In positive resist, more dose makes the opening wider. Focus decides how sharp the edge is."
- **ASK:** "If I double the spin speed, does the film become half as thick?" **Expected:** no; about 70 % of the thickness (one over the square root of two).
- **WATCH:** students treating bake temperature as a minor detail. Ask what the resist does if solvent stays in it.

[30–60] Guided run: cues for each step.

Step	What students do	What to say
1. Enter the bay	Gown, gloves, yellow-light check	"Particles and white light both ruin resist. Gown first, always."

Step	What students do	What to say
2. Prime	Inspect wafer, exhaust check, load, set recipe, run	"HMDS makes the surface water-repellent so resist sticks. The exhaust check comes before the oven runs."
3. Coat	Bottle check, exhaust, vacuum, set recipe, dispense, spin, remove	"Check the bottle date and lot. Dispense at the centre; the spin spreads it."
4. Soft bake	Set, wait for temperature, place with tweezers, time, remove, cool	"Wait until the plate is at temperature. Hands never touch the plate."
5. Expose	Interlock, mask, align, set dose and focus, expose, unload	"The interlock check is a safety step. Dose and focus are two different knobs."
6. Post-exposure bake	Set, place, time, remove, cool	"A short bake smooths the interference ripples."
7. Develop	Apron and goggles, bath check, set, develop, rinse, dry, waste	"The developer is corrosive: protection first. Waste goes in the labelled container."
8. Hard bake	Set, place, time, remove	"Too hot and the pattern flows and rounds off."
9. Inspect and decide	Inspect, measure, then Pass or Rework	"The decision is scored. If a number is out of specification, release is wrong."

[60–85] What-if exercises (Practice mode). Students predict first, change one setting, then compare.

Change	Result in the trainer	Teaching point
Dose 150 → 200 mJ/cm²	Space width 1.27 µm, over-exposed	More dose widens positive-resist openings
Spin 4,000 → 3,000 rpm	Thickness 1.15 µm	Slower spin gives a thicker film
Soft-bake 100 → 130 °C	Space width 0.82 µm, resist less sensitive	A hot bake acts like under-exposure
Focus 0 → 1.0 µm	Thickness and width pass, but defocus is flagged	A passing number does not prove a good profile

[85–110] Assessment run. Each student performs the full SOP in Assessment mode.

[110–120] Debrief.

Question	Model answer
Dose is 10 % high. What should you check first on the stepper?	The dose setting and the log; but also bake temperatures, because they shift effective dose
Why does a CD pass but the report still flags defocus?	Linewidth is one measure; sidewall slope needs its own check
Which 6D simulator session matches this lesson?	Front-End STEP 03 Photolithography and its Fault Case (Donut pattern)

3. Etching SOP Lesson (Oxide RIE)

Students etch 500 nm of silicon dioxide through a resist mask using a fluorine plasma, then strip the resist and inspect. Pair this lesson with Front-End STEP 04 Etching in the 6D simulator curriculum.

Objectives. Students can (1) perform the etch SOP including load-lock and plasma safety steps, (2) calculate etch time with over-etch, (3) explain how pressure, power and time change depth, wall angle and silicon loss.

Standard recipe and specification

Setting	Standard value	Tolerance
RF power	150 W	± 20
Chamber pressure	50 mTorr	± 10
CF_4 gas flow	40 sccm	± 5
Etch time	330 s	± 30

Specification: oxide etched at least 500 nm; silicon lost at most 10 nm; sidewall angle at least 85°; resist remaining at least 50 %. Standard settings give 550 nm, 3.3 nm, 88° and 95 %.

Script

[0–10] Safety hook.

- **ASK:** "Etch gases are invisible and some are toxic. What tells you the room is safe?" **Expected:** exhaust and gas cabinet alarm panels, the status board, and trained response; never your nose.
- **SAY:** "In this trainer, checking alarms is a safety step. Skipping it, or doing it after the wafer is loaded, loses safety marks."

[10–30] Concept script.

- **SAY:** "Lithography drew the pattern in resist. Etching cuts it into the film below. Reactive ion etching uses a plasma: chemistry removes the oxide and ions hitting the surface keep the walls straight."
- **SAY:** "Three numbers describe a good etch. Etch rate is depth per minute. Selectivity is how much faster the oxide etches than the mask or the silicon. Sidewall angle tells you how vertical the wall is."

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ER = Δx / t          S = ER_target / ER_mask
```

- **DO:** Worked example on the board. 500 nm at 100 nm/min takes 5.0 min. Add 10 % over-etch: 5.5 min, which is 330 s, the standard recipe time.
- **ASK:** "Why add over-etch?" **Expected:** film thickness and rate vary across the wafer, so the thickest area must clear.
- **WATCH:** students who add a lot of extra time "to be safe". Extra time costs silicon below the oxide.

[30–60] Guided run: cues for each step.

Step	What students do	What to say
1. Enter the bay	Gown, gloves, status board	"Tool status first: no alarms, no run."

Step	What students do	What to say
2. Check wafer and tool	Inspect, read run sheet, check alarms, check conditioning log	"The run sheet tells you the film and its thickness. Your etch time comes from that number."
3. Load	Vent, place, pump, transfer	"Vent, load, pump, transfer, in that order. Never bypass the interlocks."
4. Set recipe and etch	Select, set, door interlock, start, endpoint, plasma off	"Confirm the door is interlocked before plasma on. Watch the endpoint signal; it tells you when the oxide has cleared."
5. Unload and strip	Unload, ash, rinse in the hood, dry	"Solvent rinses happen in the fume hood with gloves."
6. Inspect and decide	Inspect, measure, decide	"Four numbers decide: depth, silicon loss, wall angle, resist left."

[60–85] What-if exercises (Practice mode).

Change	Result in the trainer	Teaching point
Etch time 330 → 240 s	400 nm etched; about 100 nm of oxide left in the openings	Under-etch leaves residue that blocks the next process
Etch time 330 → 600 s	1,000 nm etched; 33 nm of silicon lost	Over-etch eats the layer below because selectivity is finite
Pressure 50 → 120 mTorr	Wall angle 81°; etched depth falls to 462 nm	High pressure makes the etch more isotropic and slower
RF power 150 → 300 W with time 165 s	Same 550 nm depth, passes specification, but settings are out of tolerance	Different recipes can reach the same result; the SOP tolerance protects repeatability

[85–110] Assessment run. Each student performs the full SOP in Assessment mode.

[110–120] Debrief.

Question	Model answer
800 nm of oxide at 160 nm/min with 10 % over-etch: how long?	5 min + 0.5 min = 5.5 min
Why does high power lower selectivity to the resist?	More ion bombardment attacks the mask as well
Which 6D simulator session matches this lesson?	Front-End STEP 04 Etching and its Fault Case (Edge pattern)

4. Ion Implantation and Anneal SOP Lesson

Students implant arsenic into a patterned wafer, anneal it, and check sheet resistance and junction depth. The implant time is not given: students calculate it from the target dose. Pair this lesson with Front-End STEP 06 Ion Implantation in the 6D simulator curriculum.

Objectives. Students can (1) perform the implant and anneal SOP with gas and high-voltage safety checks, (2) calculate implant time from dose, beam current and wafer area, (3) explain how energy, tilt, time and anneal change dose, resistance and junction depth.

Standard recipe and specification

Setting	Standard value	Tolerance
Implant energy	50 keV	± 5
Wafer tilt	7°	± 2
Implant time	28 s (calculated)	± 4
Anneal temperature	1,000 °C	± 25
Anneal time	10 s	± 3

Fixed by the run sheet: arsenic; target dose 1.0×10^{15} ions/cm²; beam current 1.0 mA; 150 mm wafer, area about 177 cm².

Specification: delivered dose $1.0 \pm 0.1 \times10^{15}$ ions/cm²; sheet resistance 100–140 Ω/sq; junction depth at most 100 nm. Standard settings give 0.99, 121 Ω/sq and 74 nm.

Script

[0–10] Safety hook.

- **ASK:** "The dopant gases here are among the most toxic materials in the building. What do you do if the gas alarm sounds?" **Expected:** leave the area, alert others, call for help; do not investigate.
- **SAY:** "The trainer scores the gas and high-voltage checks as safety steps. They must happen before the wafer is loaded."

[10–30] Concept script.

- **SAY:** "An ion implanter fires charged dopant atoms into silicon. Two knobs matter. Energy decides how deep the ions go. Dose decides how many arrive per square centimetre."
- **SAY:** "With a fixed beam current, dose follows the charge delivered: current times time, divided by the electron charge and the wafer area."

```
Q = I · t / (q · A)          t = Q · q · A / I
```

- **DO:** Worked example on the board. Q = 1×10^{15} ions/cm², q = 1.602×10^{-19} C, A = π × 7.5² = 176.7 cm², I = 1.0 mA. Then t = 1×10^{15} × 1.602×10^{-19} × 176.7 / 0.001 ≈ 28 s.
- **SAY:** "The implant wrecks the crystal. A short, hot anneal repairs it and puts the dopants where they conduct. Too cool and they stay inactive; too hot or too long and they spread deeper. We also tilt the wafer about 7° so ions do not slip down open crystal lanes, which is called channeling."
- **ASK:** "What do energy and dose each control?" **Expected:** energy controls depth; dose controls concentration.
- **WATCH:** students who enter 28 s without understanding it. Ask them to reproduce the calculation without notes.

[30–60] Guided run: cues for each step.

Step	What students do	What to say
1. Enter the bay	Gown, gloves, status board	"No alarms, no run."

Step	What students do	What to say
2. Check wafer and implanter	Inspect, read run sheet, gas alarms, panels and interlocks	"The run sheet gives dopant, energy, dose and beam. Interlocks are closed before the beam is on."
3. Load	Vent, place, pump, transfer	"Same load-lock rule as every vacuum tool."
4. Set recipe and implant	Set, Faraday cup, start, dose counter, unload	"Check the beam current before you start, because your time assumes 1.0 mA."
5. Strip, clean and anneal	Strip, set, load, run, cool and unload	"Use tweezers: the chamber and wafer are very hot."
6. Measure and decide	Probe, measure, decide	"Sheet resistance tells you how many dopants are active; depth tells you how far they spread."

[60–85] What-if exercises (Practice mode).

Change	Result in the trainer	Teaching point
Implant time 28 → 20 s	Dose 0.71×10^{15}; sheet resistance 170 Ω/sq	Too few ions: resistance rises
Energy 50 → 80 keV	Junction depth 114 nm	Higher energy goes deeper
Tilt 7° → 0°	Channeling flagged; depth 97 nm	Channeling is a risk even when the depth number looks fine
Anneal 1,000 → 900 °C	Sheet resistance 162 Ω/sq; junction 34 nm	A cool anneal keeps the junction shallow but leaves dopants inactive

[85–110] Assessment run. Each student performs the full SOP in Assessment mode.

[110–120] Debrief.

Question	Model answer
Beam current drops to 0.5 mA. What happens to dose at 28 s?	It halves; time must double to 56 s
Why can a wafer pass on depth yet fail the report?	Channeling and other process flags count even when one number is within limit
Which 6D simulator session matches this lesson?	Front-End STEP 06 Ion Implantation and its Fault Case (Donut pattern)

5. CVD Deposition SOP Lesson (PECVD Oxide)

Students deposit 500 nm of silicon dioxide by plasma-enhanced CVD and check thickness, uniformity and refractive index. Pair this lesson with Front-End STEP 05 Deposition in the 6D simulator curriculum.

Objectives. Students can (1) perform the PECVD SOP with silane-safety and load-lock steps, (2) calculate deposition time from growth rate, (3) explain how temperature, pressure, power and gas flow change thickness, uniformity and film quality.

Standard recipe and specification

Setting	Standard value	Tolerance
Deposition temperature	350 °C	± 25
Chamber pressure	1.0 Torr	± 0.2
RF power	200 W	± 25
SiH_4 flow	50 sccm	± 5
Deposition time	150 s	± 10

Specification: thickness 500 ± 30 nm; non-uniformity at most 3.0 %; refractive index 1.46 ± 0.02. Standard settings give 500 nm, 1.5 % and 1.460.

Script

[0–10] Safety hook.

- **ASK:** "Silane ignites by itself in air. What does that tell you about how we treat the gas cabinet and the exhaust scrubber?" **Expected:** we verify alarms and scrubber before any run, and never open a gas line.
- **SAY:** "The silane check is a safety step. It must be done before you load the wafer."

[10–30] Concept script.

- **SAY:** "CVD adds a film on top of the wafer without consuming it. Gases flow in, the plasma and heat make them react, and a solid layer builds up on the surface. In PECVD the plasma supplies energy, so we can deposit at a lower temperature, which protects metal already on the wafer."
- **SAY:** "Growth rate is thickness divided by time. At the standard settings it is about 200 nm per minute, so 500 nm takes 2.5 minutes, which is 150 seconds."
- **SAY:** "Pressure and power shape the plasma, so they change how even the film is. Silane flow changes the film composition, which shows up as refractive index. A cool wafer gives a porous, low-density film."
- **ASK:** "The film is 20 % too thin. Name two ways to fix it." **Expected:** longer time, or higher growth rate (power, silane flow); then check uniformity and refractive index, because rate changes can affect them.
- **WATCH:** students who fix thickness only by time and ignore the other two measurements.

[30–60] Guided run: cues for each step.

Step	What students do	What to say
1. Enter the bay	Gown, gloves, status board	"No alarms, no run."

Step	What students do	What to say
2. Check chamber and gases	Clean log, gas alarms and scrubber, run sheet	"A dirty chamber sheds particles into your film. Check the log."
3. Load	Vent, place, pump, transfer	"Vent, load, pump, transfer."
4. Set recipe and deposit	Select, set, interlock, stabilise, start, plasma off, purge	"Wait for temperature and pressure to stabilise before plasma on."
5. Unload and cool	Transfer, vent, remove with tweezers, cool	"The wafer and carrier are hot. Tweezers and the cool station."
6. Measure and decide	Look, measure, decide	"Three numbers decide: thickness, uniformity, refractive index."

[60–85] What-if exercises (Practice mode).

Change	Result in the trainer	Teaching point
Pressure 1.0 → 1.5 Torr	Thickness 525 nm passes; non-uniformity 3.5 % fails	A film can pass thickness and still fail uniformity
SiH_4 50 → 60 sccm	Thickness 579 nm; refractive index 1.490	More silane: faster, and silicon-rich
Time 150 → 180 s	Thickness 600 nm	Thickness scales with time
Temperature 350 → 280 °C	Thickness 472 nm passes; porous film flagged	A passing number can hide poor film quality

[85–110] Assessment run. Each student performs the full SOP in Assessment mode.

[110–120] Debrief.

Question	Model answer
Growth rate 200 nm/min: how long for 650 nm?	3.25 min, which is 195 s
Why use PECVD instead of a hot furnace deposition?	Lower temperature protects metal layers already on the wafer
Which 6D simulator session matches this lesson?	Front-End STEP 05 Deposition and its Fault Case (Edge pattern)

6. Metallization SOP Lesson (Aluminium Sputter)

Students sputter 1 µm of aluminium over a titanium/titanium-nitride barrier and check thickness, sheet resistance, uniformity and contact resistance. The Ti/TiN barrier is deposited in an earlier step and is not part of this recipe. Pair this lesson with Front-End STEP 07 Metallization in the 6D simulator curriculum.

Objectives. Students can (1) perform the sputter SOP including target, vacuum, cooling and interlock checks, (2) relate resistivity, thickness and sheet resistance, (3) explain why the wafer pre-clean, the target pre-sputter, pressure and substrate temperature matter.

Standard recipe and specification

Setting	Standard value	Tolerance
Wafer pre-clean time (Ar sputter etch, shutter open)	30 s	± 10
Target pre-sputter time (shutter closed)	30 s	± 10
Argon flow	30 sccm	± 5
Process pressure	3 mTorr	± 1
DC power	5.0 kW	± 0.5
Substrate temperature	200 °C	± 25
Deposition time	100 s	± 5

Specification: thickness 1,000 ± 50 nm; sheet resistance at most 33 mΩ/sq; non-uniformity at most 4 %; contact resistance (an illustrative training value) at most 1.5 × the standard value. Standard settings give 1,000 nm, 29.0 mΩ/sq, 2.0 % and 1.0 ×.

Script

[0–10] Safety hook.

- **ASK:** "The sputter source has high voltage, strong magnets and flowing cooling water, all in one place. Which of these is easy to forget when you are focused on the wafer?" **Expected:** accept any answer with a reason; stress keeping hands and tools clear of the source and confirming the shields and interlocks.
- **SAY:** "Shield and interlock checks are scored as safety steps. A wafer in the chamber is not a reason to skip them."

[10–30] Concept script.

- **SAY:** "In sputtering, argon ions strike a metal target and knock atoms loose. Those atoms travel to the wafer and stick. The wafer is heated to help the film form well."
- **SAY:** "Before depositing, two different clean steps run. The wafer pre-clean is an argon sputter etch: the shutter is open, the substrate is biased, and the argon ions remove the thin native oxide so the metal bonds to the surface underneath. It often runs in its own chamber, and the wafer moves on under vacuum."
- **SAY:** "The target pre-sputter is a different job. With the shutter closed over the wafer, the plasma cleans the oxide and contamination off the target. The shutter keeps that material off the wafer, so a closed shutter can never clean the wafer."
- **SAY:** "A thin film is described by sheet resistance: resistivity divided by thickness. Aluminium is about 2.9 microohm-centimetres, so a 1 micrometre film is about 0.029 ohms per square."

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Rs = ρ / t          R = Rs · L / W
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- **DO:** Worked example. A line of 100 squares at 29 mΩ/sq has a resistance of 2.9 Ω. If the film is half as thick, resistance doubles.

• **ASK:** "Why is a short wafer pre-clean a risk for a chip, even if the film thickness and sheet resistance are perfect?" **Expected:** the interface keeps native oxide, so adhesion is poor and contact resistance is high. The film itself is unchanged, so its sheet resistance does not show it.

• **ASK:** "The shutter is closed during the target pre-sputter. Why?" **Expected:** the oxide and contamination on the target sputter off, and the shutter keeps them away from the wafer.

• **WATCH:** students who read only thickness, or who believe sheet resistance shows a dirty interface. Point to the four-number rule: thickness, sheet resistance, uniformity and contact resistance.

[30–60] Guided run: cues for each step.

Step	What students do	What to say
1. Enter the bay	Gown, gloves, status board	"No alarms, no run."
2. Check target, vacuum, cooling	Target ID and life, base pressure, water and argon, shields	"A poor vacuum puts oxygen and water into your metal. Check it before loading."
3. Load	Vent, place, pump, transfer	"Vent, load, pump, transfer."
4. Set recipe and sputter	Select, set, interlock, heat, wafer pre-clean, target pre-sputter, deposit, plasma off	"Pre-clean the wafer with the shutter open. Then close the shutter and condition the target. Then open the shutter and deposit."
5. Unload and cool	Transfer, vent, remove with tweezers, cool	"The wafer is hot. Tweezers and the cool station."
6. Measure and decide	Look, measure, test contact resistance, decide	"Four numbers decide: thickness, sheet resistance, uniformity, contact resistance."

[60–85] What-if exercises (Practice mode).

Change	Result in the trainer	Teaching point
Wafer pre-clean 30 → 10 s	Sheet resistance stays 29.0 mΩ/sq; contact resistance 2.0 × standard (illustrative); adhesion risk flagged	Native oxide hurts the interface, not the film
Target pre-sputter 30 → 10 s	Sheet resistance 30.5 mΩ/sq (illustrative), still in specification; target-conditioning warning	A dirty target puts impurities into the film
Pressure 3 → 6 mTorr	Thickness 910 nm; 37.6 mΩ/sq; non-uniformity 5.0 %	One knob can break three measurements
Substrate 200 → 400 °C	Sheet resistance falls to 25.5 mΩ/sq; hillocks flagged	Lower resistance is not always better: bumps can short layers
DC power 5.0 → 4.0 kW	Thickness 800 nm; 36.3 mΩ/sq	Rate scales with power; a thin film also raises resistance

[85–110] Assessment run. Each student performs the full SOP in Assessment mode.

[110–120] Debrief.

Question	Model answer
Rate is 600 nm/min: time for 1,200 nm?	2 min, which is 120 s
Film thickness halves: effect on sheet resistance?	It doubles
Wafer pre-clean is skipped. Which number changes: sheet resistance or contact resistance?	Contact resistance. The sheet resistance of the film is unchanged.
Which 6D simulator session matches this lesson?	Front-End STEP 07 Metallization and its Fault Case (Edge pattern)

7. Back-End SOP Lessons (Assembly and Test)

These five lessons let students perform the assembly and test steps of a package line in the order the 6D back-end simulator teaches them: die bonding, wire bonding, molding, saw/dicing and package testing. Each uses the same 120-minute lesson model, scoring and work orders as Parts 2–6. Settings are typical of a teaching lab with semi-automatic bench equipment; replace them with the partner lab's real equipment data (Appendix). Back-end students handle bare die, so every lesson starts with ESD or heat and press safety.

7.1 Die Bonding (Epoxy Die Attach)

Students attach a die to a lead frame with epoxy and cure it, then check bond-line thickness, voids and shear strength. Pair with Back-End MODULE 01 Die Bonding in the 6D simulator curriculum.

Objectives. Students can (1) prepare an ESD-safe bench and check epoxy, frame and die, (2) scale epoxy dispense time and bond force to die size, (3) explain how force, volume and cure change thickness, voids and strength.

Standard recipe (Standard work order, 2.0 mm die).

Setting	Standard value	Tolerance
Epoxy dispense time	120 ms	± 20
Bond force	100 gf	± 20
Cure temperature	150 °C	± 10
Cure time	60 min	± 10

Specification: bond-line thickness 25 ± 5 µm; void area at most 5 %; die shear strength at least 2.4 kgf. Standard settings give 25.0 µm, 1.5 % and 3.00 kgf.

[0–10] Safety hook.

- **ASK:** "You feel a static spark only when it is thousands of volts. Many chips are damaged at far lower. What does that tell you about the wrist strap?" **Expected:** you cannot feel damaging static, so you must always be grounded.
- **SAY:** "The strap test and the exhaust check are scored as safety steps. If you skip them, you lose safety marks even if the die survives."

[10–30] Concept script.

- **SAY:** "Die bonding glues the die to the lead frame. The glue holds the die, carries heat away, and for some devices carries current. Epoxy volume sets how thick the glue layer is. Bond force presses air out of it. Air bubbles, called voids, slow the heat and weaken the bond."
- **SAY:** "Epoxy volume and force both scale with die area. A 3.0 mm die has 2.25 times the area of a 2.0 mm die, so it needs about 2.25 times the dispense time and force: 270 ms and 225 gf."
- **SAY:** "The cure is a time and temperature trade-off. A warmer oven cures faster, but you stay inside the epoxy datasheet window."
- **WATCH:** students who add extra force "to be safe". Too much force squeezes the layer thin and cracks the die.

[30–60] Guided run: cues for each step.

Step	What students do	What to say
1. Prepare the ESD bench	Smock, wrist strap test, mat check	"Test the strap every time. A broken strap protects nothing."

Step	What students do	What to say
2. Check the materials	Epoxy label, exhaust, frame, wafer ring	"Epoxy has a working life once thawed. The label tells you."
3. Dispense and place	Clamp, set recipe, guard, dispense, pick, place, check	"Dispense in the centre. The fillet should show on all four sides."
4. Cure	Set, load, timer, gloves, remove, cool	"Put on the heat gloves before you open a hot oven."
5. Inspect and decide	Inspect, measure, decide	"Three numbers decide: thickness, voids, shear."

[60–85] What-if exercises (Practice mode, Standard work order).

Change	Result in the trainer	Teaching point
Bond force 100 → 60 gf	Bond line 32.3 µm; voids 5.1 %	Light force leaves air in the epoxy
Dispense 120 → 200 ms	Bond line 37.6 µm	More epoxy gives a thicker layer
Cure 150 → 130 °C	Die shear 1.10 kgf	A cool cure leaves a weak bond
Bond force 100 → 200 gf	Bond line 17.7 µm; die-crack risk flagged	Too much force is as bad as too little

[85–110] Assessment run. Each student performs the SOP in Assessment mode, ideally on a different work order.

[110–120] Debrief.

Question	Model answer
A 1.5 mm die: what dispense time and force?	About 68 ms and 56 gf (area is 0.56 of the 2.0 mm die)
Why do voids matter?	They slow heat flow and weaken the bond
Which later station can an epoxy mistake affect?	Wire bonding, if epoxy squeezes onto the bond pads

7.2 Wire Bonding (Gold Ball Bonding)

Students set up a wire bonder, teach the bond points, bond a lead frame, and test the bonds. Pair with Back-End MODULE 02 Wire Bonding in the 6D simulator curriculum.

Objectives. Students can (1) set up wire, capillary and frame safely, (2) scale force and power to the wire diameter, (3) explain how force, power, time and heat change bond strength, ball size and loop.

Standard recipe (Standard work order, 25 µm wire).

Setting	Standard value	Tolerance
Bond force	50 gf	± 10
Ultrasonic power	80 mW	± 10
Bond time	15 ms	± 3
Stage temperature	175 °C	± 15
Loop height	150 µm	± 20

Specification: ball shear at least 36 gf; wire pull at least 6 gf; ball diameter 55–70 µm; loop height within ± 20 % of the recipe. Standard settings give 45.0 gf, 9.0 gf, 60 µm and 150 µm.

[0–10] Safety hook.

- **ASK:** "The heater block is at 175 °C and a spark makes the ball. Where do your hands go during the cycle?" **Expected:** away from the stage, with the guard closed.
- **SAY:** "Closing the guard is a scored safety step, and it must come before the test bond."

[10–30] Concept script.

- **SAY:** "A spark melts the tip of the gold wire into a ball. The capillary presses the ball onto the die pad with force, ultrasonic vibration and heat, and the ball welds. That is the first bond. The machine draws the wire into a loop and presses it onto the lead finger: the stitch, or second bond."

• **SAY:** "Force and power scale with the wire cross-section, so with the diameter squared. An 18 µm wire needs 26 gf and 41 mW, not 50 and 80. Loop height is about 6 times the wire diameter, which is 108 µm."

• **SAY:** "We test two ways: shear the ball sideways, and pull the loop upward. A pull that breaks the wire is good; a bond that lifts off the pad is bad."

• **ASK:** "Why is more power not better?" **Expected:** the ball flattens and can crack the silicon under the pad (cratering).

• **WATCH:** students who copy the 25 µm settings onto a thinner wire.

[30–60] Guided run: cues for each step.

Step	What students do	What to say
1. Prepare the ESD bench	Smock, strap test, mat	"Same ESD rule as die bonding."
2. Set up wire and frame	Spool, capillary, mount frame, cover	"A worn or dirty capillary makes weak bonds."
3. Teach bond points	First bond, second bond, wire path	"The path must clear the die edge, or the wire shorts."
4. Set recipe and bond	Set, guard, test bond, look, run	"Always bond a sacrificial unit first and look at it."
5. Test and decide	Inspect, shear and pull test, decide	"Four numbers decide: shear, pull, ball size, loop."

[60–85] What-if exercises (Practice mode, Standard work order).

Change	Result in the trainer	Teaching point
Force 50 → 30 gf	Ball 53 µm, below the 55 µm limit	Light force makes a small, weak ball
Power 80 → 160 mW	Ball 71 µm; cratering risk flagged	Too much power flattens the ball
Loop 150 → 90 µm	Loop out of specification	A low loop can touch the die edge
Stage 175 → 120 °C	Ball shear 35.1 gf, below the limit	A cool stage gives poor sticking

[85–110] Assessment run. Each student performs the SOP in Assessment mode.

[110–120] Debrief.

Question	Model answer
20 µm wire: force, power and loop?	About 32 gf, 51 mW and 120 µm
Which later station is affected by a high loop?	Molding: high loops are swept by the compound
How does this link to the 6D simulator?	Back-End MODULE 02 Wire Bonding Fault Case (Scratch pattern): the same scratch shape can also come from dicing; only station data separates them

7.3 Molding (Transfer Molding)

Students mold a wire-bonded strip in epoxy moulding compound (EMC), post-cure it, and check fill, wire sweep, voids and cure. Pair with Back-End MODULE 03 Molding in the 6D simulator curriculum.

Objectives. Students can (1) run the press safely, (2) read the EMC datasheet and set the recipe, (3) explain how temperature, pressure and transfer time change fill, sweep and voids.

Standard recipe (Standard work order, EMC-A).

Setting	Standard value	Tolerance
Mold temperature	175 °C	± 5
Transfer pressure	7 MPa	± 1
Transfer time	12 s	± 3
Cure time	90 s	± 10

Specification: cavity fill at least 99 %; wire sweep at most 4 %; void area at most 3 %; cure degree at least 90 %. Standard settings give 100 %, 2.5 %, 0.8 % and 100 %.

[0–10] Safety hook.

- **ASK:** "The mold is at 175 °C and the press closes with high force. Name two things you must do before the press starts." **Expected:** heat gloves on, and the guard closed with the emergency stop clear.
- **SAY:** "Heat gloves and the guard check are both scored, in that order. Strips are ejected with the tool, never by hand."

[10–30] Concept script.

- **SAY:** "EMC is epoxy filled with silica. A pellet melts in a pot, a plunger pushes it through runners into the cavities, and the heat cures it. The compound starts to gel as soon as it is hot, so there is a limited time to fill."
- **SAY:** "Four defects to know. Short shot: the cavity is not full. Wire sweep: flow bends the wires. Voids: trapped air. Under-cure: the package stays soft. Values on the datasheet balance all four."

```
Sweep (%) = wire displacement / wire length × 100
```

- **ASK:** "Why does a hotter mold risk an incomplete fill?" **Expected:** the compound gels sooner, so the same transfer time is too long.
- **WATCH:** students who shorten transfer time to "speed up". Fast flow sweeps the wires.

[30–60] Guided run: cues for each step.

Step	What students do	What to say
1. Protective equipment	Glasses and smock, dust mask and gloves, find the stop	"EMC dust irritates. Know where the stop is."
2. Check the materials	EMC label, mold surface, strips	"A strip with a loose wire will not survive the flow."
3. Set the mold recipe	Set, wait for temperature	"All four values come from the datasheet."
4. Load	Heat gloves, strips, pellets	"Everything in the open mold is very hot."
5. Run the press	Guard check, start, watch, open	"Watch pressure and time; they are your evidence if something goes wrong."
6. Eject and post-cure	Eject tool, deflash, oven rack, timer	"Not by hand. Ever."
7. Inspect and decide	Inspect, measure, decide	"Four numbers decide: fill, sweep, voids, cure."

[60–85] What-if exercises (Practice mode, Standard work order).

Change	Result in the trainer	Teaching point
Pressure 7 → 4 MPa	Voids 3.8 %	Low pressure leaves air in the compound
Transfer time 12 → 6 s	Wire sweep 5.7 %	Fast flow bends the wires
Mold temperature 175 → 190 °C	Cavity fill 94 % (short shot)	A hot mold gels before it fills
Cure time 90 → 70 s	Cure degree 78 %	Short cure leaves a soft package

[85–110] Assessment run. Each student performs the SOP in Assessment mode on a different EMC datasheet.

[110–120] Debrief.

Question	Model answer
Wire 3.0 mm long moves 0.12 mm. Sweep?	4 %
Which earlier station affects sweep?	Wire bonding: a high loop is swept more
Which 6D simulator session matches this lesson?	Back-End MODULE 03 Molding and its Fault Case (Center pattern)

7.4 Saw / Dicing (Singulation)

Students mount a molded strip on tape, cut it into units with a diamond blade, and check chipping, cut depth and scratches. In this flow the cut separates the molded units, as in the 6D simulator. Pair with Back-End MODULE 04 Saw/Dicing.

Objectives. Students can (1) check the blade and the saw interlock, (2) read the blade datasheet and set the recipe, (3) explain how feed, spindle speed, coolant and depth cause chips, scratches and incomplete cuts.

Standard recipe (Standard work order).

Setting	Standard value	Tolerance
Spindle speed	30 krpm	± 3
Feed rate	50 mm/s	± 10
Coolant flow	1.5 L/min	± 0.3
Cut depth below the package bottom	50 µm	± 15

Specification: edge chipping at most 25 µm; cut depth into the tape between 0.4 and 2 times the recipe value (20–100 µm here); no scratched units. Standard settings give 15 µm chipping and no scratches.

[0–10] Safety hook.

- **ASK:** "The blade turns at tens of thousands of revolutions per minute. When may you open the door?" **Expected:** only when the spindle has fully stopped.
- **SAY:** "The door interlock check and the waste container are scored safety steps."

[10–30] Concept script.

- **SAY:** "A thin diamond blade cuts through the molded strip and slightly into the tape underneath. If it does not reach the tape, units stay joined. If it cuts through the tape, units scatter."
- **SAY:** "Chipping grows with feed rate and shrinks with spindle speed. Coolant cools the blade and washes debris away. Too little coolant and the debris drags across the surface and scratches it."
- **ASK:** "A strip comes out with scratches. Name two stations that could have caused them." **Expected:** the saw (low coolant, debris) and wire bonding (capillary contact); only station data tells them apart.
- **WATCH:** students who raise the feed rate to finish sooner. Chipping rises quickly.

[30–60] Guided run: cues for each step.

Step	What students do	What to say
1. Protective equipment	Glasses, gloves, water check	"Water and debris fly. Eyes first."
2. Check blade and saw	Blade log, nozzles, door interlock	"A worn blade chips. The log tells you its age."
3. Mount and align	Tape, chuck, alignment	"If the alignment is off, the blade cuts through the units."
4. Set recipe and cut	Set, first cut, inspect, remaining cuts	"Stop after the first cut and look. Do not trust the whole strip to a recipe."
5. Rinse and unload	Rinse, frame, waste	"Wash debris off before it dries on."
6. Inspect and decide	Inspect, measure, decide	"Three numbers decide: chipping, depth, scratches."

[60–85] What-if exercises (Practice mode, Standard work order).

Change	Result in the trainer	Teaching point
Feed 50 → 120 mm/s	Chipping 30 µm	Fast feed breaks the edges
Spindle 30 → 12 krpm	Chipping 26 µm	Slow spindle chips more
Coolant 1.5 → 0.8 L/min	3 of 20 units scratched	Low coolant lets debris scratch
Cut depth 50 → 15 µm	Incomplete cut: units stay joined	The blade must reach the tape
Cut depth 50 → 120 µm	Tape cut through: units scatter	Too deep cuts the tape

[85–110] Assessment run. Each student performs the SOP in Assessment mode on a different blade and package datasheet.

[110–120] Debrief.

Question	Model answer
Unit pitch 5.0 mm and kerf 0.1 mm: unit width?	4.9 mm
Why is the first cut inspected before the rest?	An error in a recipe ruins the whole strip otherwise
Which 6D simulator session matches this lesson?	Back-End MODULE 04 Saw/Dicing and its paired Scratch Fault Case

7.5 Package Testing (Final Test)

Students qualify a test setup on a correlation lot of 20 units (16 known good, 4 known bad). A good setup catches every bad unit and fails no good unit. Pair with Back-End MODULE 05 Package Testing in the 6D simulator curriculum.

Objectives. Students can (1) prepare an ESD-safe test bench and check socket and tray, (2) read a test spec and set the conditions, (3) explain test escapes and false failures and what causes each.

Standard recipe (Standard work order).

Setting	Standard value	Tolerance
Supply voltage	3.3 V	± 0.1
Test temperature	25 °C	± 5
Test time per unit	50 ms	± 10
Socket contact force	150 gf	± 30

Specification: all 4 defective units caught; no good unit wrongly failed; reported yield 80 %, the true value for this lot. Standard settings give 4 of 4, 0 and 80 %.

[0–10] Safety hook.

- **ASK:** "A good unit is failed, and a bad unit is passed. Which mistake costs more, and to whom?" **Expected:** the escape costs more: a bad unit reaches a customer. A false failure wastes a good unit.
- **SAY:** "Heat gloves for a hot socket are a scored safety step, and so is the ESD bench."

[10–30] Concept script.

- **SAY:** "Final test is the last gate. A tester applies power and signals, measures the response, and sorts the unit into a bin. We qualify a setup on a correlation lot where we already know the answer."
- **SAY:** "Four settings matter. Supply voltage: too low and marginal good units fail; too high and you stress them. Temperature: a hot test finds units that are too slow when hot, and too cool a test lets them through. Test time: too short and slow-acting defects such as leakage are not seen. Contact force: too light and the pins do not touch, which looks like a failed unit."
- **ASK:** "The reported yield is 85 %, but the true yield is 80 %. What happened?" **Expected:** a bad unit passed: a test escape.
- **WATCH:** students who read a higher yield as good news.

[30–60] Guided run: cues for each step.

Step	What students do	What to say
1. Prepare the ESD bench	Smock, strap test, mat	"Finished packages are still static-sensitive."

Step	What students do	What to say
2. Check program, socket, tray	Spec, pins, calibration, tray	"A bent pin or an old calibration gives wrong verdicts."
3. Set the test conditions	Program, set, stabilise, self-check	"Wait for the temperature. A socket that is not at temperature tests the wrong condition."
4. Test the units	Heat gloves, load, close, test all	"Pin 1 first. A reversed unit gives a false failure."
5. Review and decide	Bin summary, compare, decide	"Three numbers decide: caught, wrongly failed, reported yield."

[60–85] What-if exercises (Practice mode, Standard work order).

Change	Result in the trainer	Teaching point
Supply 3.3 → 2.9 V	3 good units fail; yield reported 65 %	A low voltage fails marginal good units
Contact force 150 → 80 gf	2 good units fail; yield 70 %	Light contact looks like a failure
Test time 50 → 25 ms	3 of 4 defects caught; yield 85 %	A short test lets a bad unit escape
Temperature 25 → 45 °C	2 good units fail; yield 70 %	A hot test fails good units
Temperature 25 → 5 °C	3 of 4 defects caught; yield 85 %	A cool test lets a slow unit escape

[85–110] Assessment run. Each student performs the SOP in Assessment mode on a different test spec.

[110–120] Debrief.

Question	Model answer
Which setting errors cause escapes, and which cause false failures?	Short time and cool temperature cause escapes; low voltage, light contact and high temperature cause false failures
Why is a correlation lot used?	The right answer is known, so the setup can be proven
Which 6D simulator session matches this lesson?	Back-End MODULE 05 Package Testing and its Fault Case (Random pattern): a calibration fault leaves no spatial shape at all

8. Integrated Assessment and Link to the 6D Simulator

After the ten lessons, run a fault-finding assessment: each student gets one hidden what-if setting and must find it from the report. This connects SOP practice to the diagnosis skills taught in the 6D simulator.

Fault-finding assessment (45 minutes)

1. **DO:** Choose one what-if from Parts 2–7 for each student, without telling them. They perform the SOP in Assessment mode, entering the chosen wrong value in the one setting.
2. **DO:** Students read their own report and the "where to look" table, then write a half-page diagnosis: which measurement failed, which setting is the suspect, and what evidence supports it.
3. **DO:** Pairs swap diagnoses and check them against the instructor key. Score with the rubric below.

Criterion	Points	Full marks when the student...
Names the failed measurement	20	Identifies every out-of-specification result
Shortlists suspect settings	30	Lists every setting that could cause it, with a physical reason
Confirms with evidence	30	Uses the report or a controlled re-run to pick one suspect
Proposes a verified fix	20	Changes one setting and shows recovery

Tolerance stack-up (talking point)

Every setting has its own tolerance, but the wafer specification is fixed. Several settings at the edge of their tolerance can add up to a failed wafer. Ask students to find a combination of in-tolerance settings that fails (for example, in CVD: time at 160 s and RF power at 225 W). The lesson: an SOP tolerance is a safe window, not a guarantee.

Link to 6D simulator sessions

Trainer process	6D simulator session	Failure shape to discuss (as in the 6D instructor guides)
Lithography	Front-End STEP 03 Photolithography	Donut: overlay and focus errors that vary by exposure zone, while the center and edge stay in specification
Etching	Front-End STEP 04 Etching	Edge: a plasma-density gradient makes the edge dies the first to show CD deviation
Ion implantation	Front-End STEP 06 Ion Implantation	Donut: a beam-scan zone under- or over-doses one radial band
CVD	Front-End STEP 05 Deposition	Edge: gas flow at the chamber periphery differs from the center
Metallisation	Front-End STEP 07 Metallization	Edge: sputter distribution (the 6D simulator shows the same shape for copper plating and CMP)

The lessons follow the patterned-wafer flow, with ion implantation before CVD, so the STEP numbers in this table are not in ascending order. The numbers are those of the Front-End and Back-End 6D simulators.

Back-end track (for example, for Vietnam). Run Parts 7.1–7.5 in line order. Students keep one logbook across the five lessons and trace a late failure back to an early setting: a high wire loop (7.2) that is swept in molding (7.3), epoxy squeezed onto pads (7.1) that makes wire bonds fail (7.2), or a scratch that could come from wire bonding or dicing (7.4).

Trainer process	6D simulator session	Failure shape to discuss (as in the 6D instructor guides)
Die bonding	Back-End MODULE 01 Die Bonding	Local Cluster: a clogged dispenser nozzle affects a run of consecutive units

Trainer process	6D simulator session	Failure shape to discuss (as in the 6D instructor guides)
Wire bonding	Back-End MODULE 02 Wire Bonding	Scratch (Line): one bonding head drifts out of specification, so every unit it bonded forms a line
Molding	Back-End MODULE 03 Molding	Center: the center cavities of a multi-cavity mold run hotter and cure differently
Saw / dicing	Back-End MODULE 04 Saw/Dicing	Scratch (Line): one dicing street develops a local blade problem; same shape as wire bonding, different cause
Package testing	Back-End MODULE 05 Package Testing	Random (no shape): an inspection calibration drift has no position dependence, so the map shows no spatial pattern

In the debrief, repeat the suite's central message: **the shape of a failure narrows the suspects, but only station-level data confirms the diagnosis.** The trainer's report gives the station-level evidence for one wafer; the 6D simulator shows how the same fault appears across a whole wafer.

Suggested course path

- **Patterned-wafer flow (3 lessons):** lithography, then etching, then ion implantation. Students keep one logbook and trace a defect from a late-stage failure back to an early-stage setting.
- **Films and wiring (2 lessons):** CVD, then metallisation, with attention to thickness, resistance and uniformity.
- **Back-end line (5 lessons):** die bonding, wire bonding, molding, saw/dicing and package testing, as in Part 7.
- **Full cycle (10 lessons plus the assessment):** all ten in order, ending with the fault-finding assessment.

Appendix: Swapping in Partner-Lab Recipe Data

Each process is stored as a recipe file. To use a partner lab's real equipment settings, export a standard recipe, edit it, and load it back through **Recipe data → Load recipe file**. The trainer checks the file and explains any problem.

Steps

1. Choose the process in the **Process** menu, open **Recipe data**, and press **Export current recipe**.
2. Edit the exported file in a text editor: replace step titles, action wording, hazard text and each parameter's `nominal` and `tol` with the lab's values and SOP wording.
3. Keep the parameter `id` values and the final measurement step (see the tables below). The outcome model reads these ids.
4. Change `meta.source` to name the real source (for example, the lab and the SOP document number) so reports show where the numbers come from.
5. Load the file, run it in Guided mode, and have a lab technician confirm the sequence and values.

What the recipe file contains

Key	Meaning
`meta`	Names, data source, note, version
`weights`	Score weights: sequence, settings, safety (sum 100)
`model.type`	`litho`, `etch`, `implant`, `cvd`, `metal`, `diebond`, `wirebond`, `mold`, `saw` or `test`; selects the outcome model and the cross-section drawing
`stations`	The stations shown in the bay and their labels
`steps`	Ordered steps; each has actions, optional safety hazard text and optional parameters. The last step must be marked `final` and contain a `measure` action
action `safety: true`	Marks a safety step; doing it late or out of order counts as a safety violation
action `type: "set"`	Where the step's parameters are entered and graded

Parameter ids the outcome model needs

Process	Required parameter ids
Lithography	`spin_rpm`, `sb_temp`, `sb_s`, `peb_temp`, `dose`, `focus`, `dev_s` (optional: `dispense_ml`, `spin_s`, `rinse_s`, `hb_temp`)
Etching	`rf_power`, `pressure`, `gas_flow`, `etch_s`
Ion implantation	`energy`, `tilt`, `implant_s`, `rta_temp`, `rta_s`
CVD	`dep_temp`, `pressure`, `rf_power`, `sih4`, `dep_s`
Metallisation	`clean_s`, `tgt_s`, `ar_flow`, `pressure`, `dc_kw`, `sub_temp`, `dep_s`
Die bonding	`dispense_ms`, `bond_force`, `cure_temp`, `cure_min`
Wire bonding	`bond_force`, `us_power`, `bond_time`, `stage_temp`, `loop_height` (the model type also reads `wire_um`)
Molding	`mold_temp`, `pressure`, `transfer_s`, `cure_s`
Saw / dicing	`spindle_krpm`, `feed_mms`, `coolant_lpm`, `cut_depth_um`
Package testing	`vdd`, `test_temp`, `test_ms`, `contact_gf`

Metallisation recipe files exported before v0.2 use a single pre_s parameter. Replace it with clean_s (wafer pre-clean) and tgt_s (target pre-sputter) before loading the file.

Work orders in a recipe file. A recipe may include a `workOrders` list. Each entry gives an `id`, a `label`, new `nominals` and `tols` for named parameters, `model` overrides (for example the target thickness), `vars` that fill `{placeholders}` in the text, and a `brief` shown on the work-order card. The first entry is the Standard order. Export a standard recipe to see the format.

Extra parameters and extra steps can be added freely; they are scored for tolerance and sequence but do not change the process result.

Limits to explain to partners

- The result calculations are a simplified training model scaled to the standard values. If a lab's equipment behaves differently (for example, a different etch rate), the model's results will not match its real wafers. Matching them requires a developer to recalibrate the model coefficients.
- Settings are scored against the recipe's nominal value and tolerance only. The model does not know a lab's own limits unless they are written into the file.
- The trainer runs on a screen. Hand-tracked VR interaction is a later front end on the same engine and recipe files.
- The contact-resistance and target-conditioning effects in the metallisation lesson are illustrative training values and are not calibrated. The Ti/TiN barrier is not modelled: it is treated as already deposited.

Pre-release checklist for partner-lab data

☐ Equipment list and model names for each process confirmed with the lab.

☐ Step sequence matches the lab's current SOP document.

☐ Every safety step the lab requires (PPE, chemical handling, gas alarms, interlocks) appears as an action marked `safety`.

☐ Nominal values and tolerances taken from the lab's recipes, with the document and date noted in `meta.source`.

☐ Specification limits (thickness, linewidth, resistance and similar) agree with the lab's acceptance criteria. Check the model's pass thresholds in the standard recipe and discuss changes with IEG.

☐ A lab technician has run the Guided mode and approved the wording.

☐ Confidential recipe data: confirm who may see the recipe file before it is shared or hosted.